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BRAKE APPARATUS FOR VEHICLE

Abstract

A brake apparatus for a vehicle may include: a drive part configured to generate driving power; a pair of pressing parts configured to receive the driving power from the drive part and to apply or release a braking force; a load transmission part installed between the pair of pressing parts and configured to transmit a pressing load of any one of the pair of pressing parts to the other pressing part of the pair of pressing parts; and a load transmission restricting part configured to selectively restrict an operation of the load transmission part by pressing the load transmission part.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] The present application is a continuation of U.S. patent application Ser. No. 17/570,582 filed Jan. 7, 2022, which claims the benefit under 35 USC § 119 of Korean Patent Application No. 10-2021-0140911 filed on Oct. 21, 2021, in the Korean Intellectual Property Office, the entire disclosures of which are hereby incorporated by reference for all purposes.

BACKGROUND

Technical Field

[0002] The present disclosure relates to a brake apparatus for a vehicle, and more particularly, to a brake apparatus for a vehicle having a pair of pistons.

Discussion of the Background

[0003] In general, a parking brake for a vehicle includes a motor and a power transmission device for operating a friction pad installed on a caliper of a disc brake apparatus during parking.

[0004] Recently, a parking brake for a vehicle having a structure for pressing a disc using a pair of left and right pistons is applied. In this case, a method of transmitting loads of the left and right pistons using a balancing gear is used to solve a problem that braking performance deteriorates due to non-uniform transmission of loads to the left and right pistons. For example, when a load is applied to the left piston first, a right ring gear is rotated by the balancing gear to synchronize load values of the right and left pistons, which makes it possible to allow the left and right pistons to simultaneously press the pad with the loads.

[0005] However, when the brake is disengaged, a piston with a low load is released first and then a piston with a high load is released by the balancing function. Therefore, the piston with the high load is sometimes incompletely released because the release time is restricted. The incomplete release of the piston causes a drag due to a residual load.

[0006] The background technology of the present disclosure is disclosed in Korean Patent Application Laid-Open No. 10-2010-0098846 (published on Sep. 10, 2010 and entitled 'Disc Brake Having Parking Function').

SUMMARY

[0007] An object of the present disclosure is to provide a brake apparatus for a vehicle, which is capable of allowing pistons to transmit a uniform load when applying a braking force and preventing the pistons from being incompletely released when releasing the braking force.

[0008] Various embodiments are directed to a brake apparatus for a vehicle, including: a drive part configured to generate driving power; a pair of pressing parts configured to receive the driving power from the drive part and to apply or release a braking force; a load transmission part installed between the pair of pressing parts and configured to transmit a pressing load of any one of the pair of pressing parts to the other pressing part of the pair of pressing parts; and a load transmission restricting part configured to selectively restrict an operation of the load transmission part by pressing the load transmission part, wherein the load transmission part comprises: a pair of ring gear parts connected to each of the pressing parts respectively; and one or more balancing gear parts disposed between the pair of ring gear parts and configured to rotate while engaging with the ring gear parts, and wherein the load transmission restricting part comprises: a support plate fixed to a rotation shaft portion that supports the one or more balancing gear parts so that the one or more

balancing gear parts is rotatable; pressing plates disposed to face the support plate and configured to come into contact with the one or more balancing gear parts; and a pressing member disposed between the support plate and the pressing plates and configured to press the pressing plates toward the one or more balancing gear parts.

[0009] The load transmission restricting part may permit rotation of the one or more balancing gear parts when the pair of pressing parts apply the braking force, and the load transmission restricting part may restrict the rotation of the one or more balancing gear parts when the pair of pressing parts release the braking force.

[0010] The load transmission restricting part may apply a rotational resistance force to the one or more balancing gear parts by pressing the one or more balancing gear parts.

[0011] The load transmission restricting part may press the one or more balancing gear parts in an axial direction.

[0012] The pressing member may be elastically deformable in a longitudinal direction thereof.

[0013] The pressing plates may comprise: a first pressing plate disposed to face one side of the one or more balancing gear parts; and a second pressing plate disposed to face another side of the one or more balancing gear parts and configured to receive a pressing force from the pressing member.

[0014] The second pressing plate may be installed to be movable in a longitudinal direction of the rotation shaft portion.

[0015] The brake apparatus may further comprise: a stopper coupled to the rotation shaft portion and configured to support the support plate.

[0016] The stopper may be installed to be movable in a longitudinal direction of the rotation shaft portion and may be configured to adjust an interval between the support plate and the pressing plates.

[0017] According to the brake apparatus for a vehicle according to the present disclosure, when the pressing load is concentrated on any one of the plurality of pressing parts when the braking force is applied, the load transmission part may transmit the pressing load to the remaining pressing part and allow the pressing parts to press the brake pad with the uniform load.

[0018] In addition, according to the brake apparatus for a vehicle according to the present disclosure, the load transmission restricting part may restrict the operation of the load transmission part when the braking force is released. Therefore, it is possible to quickly separate the pair of piston parts from the brake pad and prevent the drag caused by the incomplete release of the piston part.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0019] FIG. 1 is a perspective view schematically illustrating a configuration of a brake apparatus for a vehicle according to an embodiment of the present disclosure.

[0020] FIG. 2 is an exploded perspective view schematically illustrating the configuration of the brake apparatus for a vehicle according to the embodiment of the present disclosure.

[0021] FIG. 3 is a front view schematically illustrating the configuration of the brake apparatus for a vehicle according to the embodiment of the present disclosure.

[0022] FIG. 4 is an enlarged view schematically illustrating an installed state of a load transmission restricting part according to the embodiment of the present disclosure.

[0023] FIG. 5 is a perspective view schematically illustrating a configuration of the load transmission restricting part according to the embodiment of the present disclosure.

[0024] FIG. 6 is an exploded perspective view schematically illustrating the configuration of the load transmission restricting part according to the embodiment of the present disclosure.

[0025] FIGS. 7 to 9 are operational views schematically illustrating a process in which the brake

apparatus for a vehicle according to the embodiment of the present disclosure applies a braking force.

[0026] FIGS. **10** and **11** are operational views schematically illustrating a process in which the brake apparatus for a vehicle according to the embodiment of the present disclosure releases the braking force.

DETAILED DESCRIPTION OF THE ILLUSTRATED EMBODIMENTS

[0027] Hereinafter, a brake apparatus for a vehicle will be described below with reference to the accompanying drawings through various exemplary embodiments.

[0028] Hereinafter, an embodiment of a brake apparatus for a vehicle according to the present disclosure will be described with reference to the accompanying drawings.

[0029] Here, thicknesses of lines illustrated in the drawings, sizes of constituent elements, or the like may be exaggerated for clarity and convenience of description. In addition, the terms used below are defined in consideration of the functions in the present disclosure and may vary depending on the intention of a user or an operator or a usual practice. Therefore, the definition of the terms should be made based on the entire contents of the present specification.

[0030] In addition, in the present specification, when one constituent element is referred to as being “connected to (or coupled to)” another constituent element, the constituent elements can be “directly connected to (coupled to)” each other, and can also be “indirectly connected to (coupled to)” each other with other elements interposed therebetween. In addition, unless explicitly described to the contrary, the word “comprise (or include)” and variations such as “comprises (or includes)” or “comprising (or including)” will be understood to imply the inclusion of stated elements, not the exclusion of any other elements.

[0031] In addition, throughout the specification, the same reference numerals denote the same constituent elements. Even though the same or similar reference numerals are not mentioned or described with reference to the specific drawings, the same or similar reference numerals may be described with reference to the other drawings. In addition, even though there are parts denoted by no reference numeral in the specific drawings, the parts may be described with reference to the other drawings. In addition, the numbers, shapes, sizes, relative differences in sizes, and the like of the detailed constituent elements illustrated in the drawings of the present application are set for convenience of understanding, do not limit the embodiments, and may be variously implemented.

[0032] FIG. **1** is a perspective view schematically illustrating a configuration of a brake apparatus for a vehicle according to an embodiment of the present disclosure, FIG. **2** is a front view schematically illustrating the configuration of the brake apparatus for a vehicle according to the embodiment of the present disclosure, and FIG. **3** is an exploded perspective view schematically illustrating the configuration of the brake apparatus for a vehicle according to the embodiment of the present disclosure.

[0033] Referring to FIGS. **1** to **3**, a brake apparatus **1** for a vehicle according to an embodiment of the present disclosure includes a housing **10**, a drive part **30**, pressing parts **100** and **200**, a load transmission part **300**, and a load transmission restricting part **400**.

[0034] The housing **10** has therein a space in which the pressing parts **100** and **200**, the load transmission part **300**, and the load transmission restricting part **400** may be installed. The housing **10** may be openable and closable so that the components installed in the housing **10** may be easily replaced or managed. A specific shape of the housing **10** is not limited to a shape illustrated in FIG. **1** but may be variously designed and changed depending on shapes and sizes the pressing parts **100** and **200**, the load transmission part **300**, and the load transmission restricting part **400**.

[0035] The drive part **30** includes an electric motor configured to generate power by being supplied with electric power from the outside, and a driving gear connected to an output shaft of the electric motor and configured to be rotated by the electric motor. The driving gear may be a spur gear, a bevel gear, a worm gear, or the like capable of transmitting a rotational force of the electric motor to the pressing parts **100** and **200** and variously designed and changed within the technical spirit of

the power transmission means.

[0036] The pressing parts **100** and **200** are installed inside the housing **10**. The pressing parts **100** and **200** apply braking forces to the vehicle by receiving the power from the drive part **30** and pressing a brake pad **20** that generates contact friction with a disc (not illustrated).

[0037] The pressing part **100** or **200** is provided in plural, and the plurality of pressing parts **100** or **200** is disposed in parallel. The pressing parts **100** and **200** are symmetrically installed at left and right sides (based on FIG. 2) based on a central portion of the brake pad **20**.

[0038] The pressing part **100** or **200** according to the embodiment of the present disclosure includes a sun gear part **110** or **210**, a connection gear part **120** or **220**, a planet gear part **130** or **230**, a carrier part **150** or **250**, and a piston part **170** or **270**.

[0039] The connection gear part **120** or **220** according to the embodiment of the present disclosure includes a connection gear main body **121** or **221**, a connection wheel **122** or **222**, and a connection insertion portion **123** or **223**.

[0040] The connection gear main body **121** or **221** defines a schematic external appearance of the connection gear part **120** or **220**. The connection gear main body **121** or **221** according to the embodiment of the present disclosure may be provided in the form of an approximately circular plate.

[0041] The connection wheel **122** or **222** is formed on an outer peripheral surface of the connection gear main body **121** or **221**. The connection wheel **122** or **222** according to the embodiment of the present disclosure may have gear teeth continuously extending along the outer peripheral surface of the connection gear main body **121** or **221**. The connection wheels **122** and **222** engage with the drive part **30**, specifically, the driving gear. Therefore, the power of the drive part **30** may be transmitted to the connection gear parts **120** and **220** and rotate the connection gear parts **120** and **220**.

[0042] The connection insertion portion **123** or **223** is formed in an internal space of the connection wheel **122** or **222**. That is, the connection wheel **122** or **222** is formed outside a wall formed along the outer peripheral surface of the connection gear main body **121** or **221**. The connection insertion portion **123** or **223** is formed in the internal space of the wall on which the connection wheel **122** or **222** is formed. The connection insertion portion **123** or **223** according to the embodiment of the present disclosure is provided in the form of a concave groove. Ring gear parts **310**, more specifically, ring gear inner portions **311**, which will be described below, are respectively inserted into the connection insertion portions **123** and **223**.

[0043] The sun gear parts **110** and **210** rotate by receiving the power from the drive part **30**. According to the present embodiment, the sun gear part **110** or **210** is coupled to the connection gear part **120** or **220**. The sun gear part **110** or **210** may rotate by means of the connection gear part **120** or **220** that receives the power from the drive part **30**.

[0044] The sun gear part **110** or **210** is concentric with a rotation center of the connection gear part **120** or **220**. Therefore, when the power is transmitted to the connection gear parts **120** and **220** by a power transmission part **70**, the connection gear part **120** rotates on the same rotation axis as the sun gear part **110**, and the connection gear part **220** rotates on the same rotation axis as the sun gear part **210**. The sun gear part **110** or **210** is disposed inside an inner peripheral surface of the connection gear part **120** or **220** on which the connection insertion portion **123** or **223** is formed.

[0045] The sun gear part **110** may be integrated with the connection gear part **120**, and the sun gear part **210** may be integrated with the connection gear part **220**. Alternatively, the sun gear part **110** or **210** may be formed separately from the connection gear part **120** or **220** and integrated with the connection gear part **120** or **220** by being coupled. Since the sun gear part **110** or **210** is integrated with the connection gear part **120** or **220**, the sun gear part **110** or **210** also rotates when the connection gear part **120** or **220** rotates by receiving the power from the drive part **30**.

[0046] The sun gear part **110** or **210** according to the embodiment of the present disclosure includes a sun gear **111** or **211**, and a sun gear connection body **112** or **212**.

[0047] The sun gear connection body **112** or **212** is coupled to the connection gear main body **121** or **221**.

[0048] The sun gear **111** or **211** is formed at a central portion of the sun gear connection body **112** or **212** and has gear tooth formed on an outer peripheral surface thereof so as to engage with the planet gear part **130** or **230**.

[0049] The planet gear part **130** or **230** engages with the sun gear **111** or **211** and rotates and revolves.

[0050] The planet gear part **130** or **230** has a plurality of planet gears **131** or **231**. The present embodiment provides an example in which four planet gears **131** or **231** are disposed, but the present disclosure is not limited thereto. The number of planet gears may be three or less or five or more.

[0051] The plurality of planet gears **131** or **231** is disposed at an equal angle around a rotation center of the sun gear **111** or **211**. The plurality of planet gears **131** or **231** engages with the sun gear **111** or **211** and rotates and/or revolves when the sun gear **111** or **211** rotates.

[0052] The planet gear part **130** or **230** is coupled to the carrier part **150** or **250**. When the plurality of planet gears **131** or **231** revolves, the carrier part **150** or **250** also rotates clockwise or counterclockwise.

[0053] The carrier part **150** or **250** according to the embodiment of the present disclosure includes a carrier main body **151** or **251**, carrier rotation shafts **152** or **252**, and a carrier connecting portion **153** or **253**.

[0054] The carrier rotation shaft **152** or **252** protrudes from the carrier main body **151** or **251** toward the planet gear part **130** or **230**.

[0055] The carrier rotation shaft **152** or **252** is provided in plural, and the plurality of carrier rotation shafts **152** or **252** is equal in number to the planet gear part **130** or **230**. The carrier rotation shaft **152** or **252** is penetratively coupled to the planet gear part **130** or **230**. Therefore, the planet gear part **130** or **230** may rotate and revolve on the carrier rotation shafts **152** or **252**.

[0056] The carrier connecting portion **153** or **253** is formed on an inner peripheral surface of the carrier main body **151** or **251** and connected to a piston connecting portion **173** or **273** of the piston part **170** or **270**.

[0057] In the present embodiment, the carrier connecting portion **153** or **253** is provided in the form of a concave groove. The piston connecting portion **173** or **273** is provided in the form of a protrusion and inserted into the carrier connecting portion **153** or **253**.

[0058] On the contrary, the piston connecting portion **173** or **273** may be provided in the form of a concave groove, and the carrier connecting portion **153** or **253** may be provided in the form of a protrusion and inserted into the piston connecting portion **173** or **273**.

[0059] The carrier connecting portion **153** or **253** may be coupled to the piston connecting portion **173** or **273** by means of splines. The carrier part **150** or **250** and the piston part **170** or **270** may of course be coupled by means of screws, for example, in addition to the splines.

[0060] The piston part **170** or **270** is connected to the carrier part **150** or **250**. The piston part **170** or **270** extends or retracts while converting a rotational motion of the carrier part **150** or **250** into a rectilinear motion. The piston part **170** or **270** applies or releases the braking force by pressing or releasing the brake pad **20** while extending or retracting.

[0061] The piston part **170** or **270** according to the embodiment of the present disclosure may include a piston main body **171** or **271**, a piston shaft **172** or **272**, and the piston connecting portion **173** or **273**.

[0062] The piston main body **171** or **271** has a hollow portion therein and moves to be in contact with the brake pad **20**. The piston main body **171** or **271** may have a cylindrical shape.

[0063] The piston main body **171** or **271** is coupled to the piston shaft **172** or **272**, and the piston connecting portion **173** or **273** is formed at an end of the piston shaft **172** or **272** directed toward the carrier part **150** or **250**.

[0064] When the carrier part **150** or **250** rotates, the piston connecting portion **173** or **273** coupled to the carrier connecting portion **153** or **253** by means of the splines rotates, such that a rotational motion of the carrier part **150** or **250** is converted into a rectilinear motion of the piston part **170** or **270**.

[0065] As the piston part **170** or **270** rectilinearly moves, the piston part **170** or **270** moves toward the brake pad **20**. Therefore, the piston part **170** or **270** comes into contact with the brake pad **20** and presses the brake pad **20**, and the braking force is generated by friction between the brake pad **20** and the disc.

[0066] The load transmission part **300** is connected to the pair of pressing parts **100** and **200** and transmits a pressing load of any one of the pressing parts **100** and **200** to the other of the pressing parts **100** and **200**.

[0067] The load transmission parts **300** according to the embodiment of the present disclosure may include the pair of ring gear parts **310** and one or more balancing gear parts **320**.

[0068] The pair of ring gear parts **310** is respectively coupled to the planet gear parts **130** and **230** of the different pressing parts **100** and **200**. Inner peripheral surfaces of the pair of ring gear parts **310** may respectively engage with the planet gear parts **130** and **230**, such that the pair of ring gear parts **310** may rotate. Outer peripheral surfaces of the pair of ring gear parts **310** may indirectly engage with each other by means of the one or more balancing gear parts **320**. That is, the balancing gear parts **320** may be disposed between the pair of ring gear parts **310** and engage with the ring gear parts **310**. Each of the ring gear parts **310** may be installed between the planet gear **131** or **231** and the connection wheel **122** or **222**.

[0069] The ring gear part **310** according to the embodiment of the present disclosure includes the ring gear inner portion **311** and a ring gear outer portion **315**.

[0070] The ring gear inner portion **311** may be disposed outside the planet gear part **130** or **230** and have an inner gear portion **312** formed along an inner peripheral surface thereof so as to engage with the planet gear part **130** or **230**.

[0071] The ring gear outer portion **315** is coupled to an outer surface of the ring gear inner portion **311** and has an outer gear portion **316** formed along an outer peripheral surface thereof so as to engage with the balancing gear part **320**. The ring gear outer portion **315** may be integrated with the ring gear inner portion **311**.

[0072] When the inner gear portion **312** of the ring gear inner portion **311** installed at one side (the left side based on FIG. 3) rotates while engaging with the planet gear part **130**, the ring gear outer portion **315** integrated with the ring gear inner portion **311** also rotates in the same direction as the inner gear portion **312**.

[0073] Therefore, a rotational force of the ring gear outer portion **315** at one side is transmitted through the balancing gear parts **320** to the ring gear part **310**, particularly, the ring gear outer portion **315** disposed at the other side (the right side based on FIG. 5).

[0074] The balancing gear parts **320** rotate while engaging with the ring gear parts **310**, particularly, the outer gear portions **316** formed on the ring gear outer portions **315**, and transmit the rotational power of the ring gear part **310** disposed at the one side to the ring gear part **310** disposed at the other side.

[0075] More specifically, when pressing loads for pressing the pair of pressing parts **100** and **200**, particularly, the pair of piston parts **170** and **270** against the brake pad **20** are not uniformly applied, the balancing gear parts **320** transmit the pressing load of the piston part **170** disposed at one side to the piston part **270** disposed at the other side, thereby bringing the pair of piston parts **170** and **270** into contact with the brake pad **20** with the uniform pressing load. On the contrary, the balancing gear parts **320** may of course transmit the pressing load of the piston part **270** disposed at the other side to the piston part **170** disposed at one side.

[0076] The balancing gear part **320** according to the embodiment of the present disclosure is provided in the form of a spur gear. A central portion of the balancing gear part **320** is inserted into

a rotation shaft portion **321**, such that the balancing gear part **320** is rotatably supported by the rotation shaft portion **321**. The balancing gear part **320** rotates while engaging with the outer gear portion **316** formed along the outer peripheral surface of the ring gear outer portion **315**. However, the balancing gear part **320** may have various shapes, such as a bevel gear shape or a helical gear shape having gear teeth inclined at a predetermined angle with respect to a rotation center axis of the balancing gear part **320**, in addition to the spur gear shape.

[0077] In addition, the present disclosure is not limited to the example in which the balancing gear part **320** has a gear shape. Various modified examples may be made. For example, the balancing gear part **320** may be provided in the form of a belt connected to the pair of ring gear parts **310** and transmit the power from one pressing part **100** to the other pressing part **200**.

[0078] The balancing gear part **320** may be provided in plural. Therefore, the present disclosure is not limited to the present embodiment in which the two balancing gear parts **320** are provided. Various modified examples may be made. For example, one or three or more balancing gear parts may be provided depending on a distance between the pair of ring gear parts **310**.

[0079] The load transmission restricting part **400** selectively restricts the operation of the load transmission part **300** by pressing the load transmission part **300**. More specifically, the load transmission restricting part **400** selectively restricts the operation of the load transmission part **300** by applying a rotational resistance force with a preset magnitude to the balancing gear parts **320** by pressing the balancing gear parts **320** in the axial direction.

[0080] That is, the load transmission restricting part **400** permits the rotations of the balancing gear parts **320** when the braking force is applied such that a relatively high operating load is applied to the pressing parts **100** and **200** by the rotational resistance force. Therefore, the load transmission restricting part **400** may allow the load transmission part **300** to smoothly eliminate a load deviation between the pair of pressing parts **100** and **200** during the process of applying the braking force.

[0081] In addition, the load transmission restricting part **400** restricts the rotations of the balancing gear parts **320** when the braking force applied by the pressing parts **100** and **200** is released such that a relatively low operating load is applied to the pressing parts **100** and **200**. Therefore, the load transmission restricting part **400** may allow the pair of pressing parts **100** and **200** to quickly separate from the brake pad **20** during the process of releasing the braking force, thereby preventing the drag caused by the residual load.

[0082] In this case, a magnitude of the rotational resistance force applied by the load transmission restricting part **400** may be set to be smaller than a magnitude of the rotational force transmitted to the balancing gear parts **320** from the ring gear part **310** when the braking force is applied by the pressing parts **100** and **200**, and larger than a magnitude of the rotational force transmitted to the balancing gear parts **320** from the ring gear part **310** when the braking force applied by the pressing parts **100** and **200** is released.

[0083] FIG. **4** is an enlarged view schematically illustrating an installed state of the load transmission restricting part according to the embodiment of the present disclosure, FIG. **5** is a perspective view schematically illustrating a configuration of the load transmission restricting part according to the embodiment of the present disclosure, and FIG. **6** is an exploded perspective view schematically illustrating the configuration of the load transmission restricting part according to the embodiment of the present disclosure.

[0084] Referring to FIGS. **4** to **6**, the load transmission restricting part **400** according to the embodiment of the present disclosure includes a support plate **410**, pressing plates **420**, pressing members **430**, and stoppers **440**.

[0085] The support plate **410** is fixed to the rotation shaft portion **321** and supports the pressing members **430** to be described below. The support plate **410** according to the embodiment of the present disclosure may be provided in the form of a plate disposed in parallel with the balancing gear parts **320** and facing the balancing gear parts **320**. The support plate **410** is spaced apart from

lower surfaces of the balancing gear parts **320** at a predetermined interval in the axial direction of the balancing gear parts **320**. The rotation shaft portions **321** may be inserted into the support plate **410** so that the support plate **410** is slidable in a longitudinal direction of the rotation shaft portion **321**. The support plate **410** may be fixed in position by adjoining the stopper **440** to be described below in the state in which the rotation shaft portions **321** are inserted into the support plate **410**. [0086] The pressing plates **420** are disposed to face the support plate **410** and in contact with the balancing gear parts **320**. The pressing plates **420** are in close contact with the balancing gear parts **320** by receiving the pressing force from the pressing members **430** to be described below.

Therefore, the rotational resistance force applied to the balancing gear parts **320** by the load transmission restricting part **400** may be generated by the frictional force applied between the pressing plates **420** and the balancing gear parts **320**.

[0087] The pressing plates **420** according to the embodiment of the present disclosure include a first pressing plate **421** and a second pressing plate **422**.

[0088] The first pressing plate **421** is disposed to face one side of each of the balancing gear parts **320**. The first pressing plate **421** according to the embodiment of the present disclosure may be provided in the form of a plate disposed in parallel with upper surfaces of the balancing gear parts **320** and facing the balancing gear parts **320**. The first pressing plate **421** is fixed at a position at which a lower surface of the first pressing plate **421** is in contact with the upper surfaces of the balancing gear parts **320**. The first pressing plate **421** is in close contact with the upper surfaces of the balancing gear parts **320** by the reaction force generated when the second pressing plate **422** to be described below is in close contact with the lower surfaces of the balancing gear parts **320**. The first pressing plate **421** may be integrated with the rotation shaft portion **321**. Alternatively, the first pressing plate **421** may be fixed to the rotation shaft portion **321** by a separate fixing means, a bonding agent, or the like in the state in which the rotation shaft portion **321** is inserted into the first pressing plate **421**.

[0089] The second pressing plate **422** is disposed to face the other side of each of the balancing gear parts **320** and receives the pressing force from the pressing members **430**. The second pressing plate **422** according to the embodiment of the present disclosure may be provided in the form of a plate disposed between the upper surface of the support plate **410** and the lower surfaces of the balancing gear parts **320**. The second pressing plate **422** is disposed in parallel with the upper surface of the support plate **410** and the lower surfaces of the balancing gear parts **320**. The second pressing plate **422** may be installed to be movable in the longitudinal direction of the rotation shaft portion **321** in the state in which the rotation shaft portion **321** is inserted into the second pressing plate **422**. The lower surface of the second pressing plate **422** adjoins the pressing members **430** to be described below and receives the pressing force from the pressing members **430**. The upper surface of the second pressing plate **422** is in close contact with the lower surfaces of the balancing gear parts **320** by the pressing force transmitted from the pressing members **430**.

[0090] Therefore, since the first and second pressing plates **421** and **422** are in close contact with the two opposite surfaces of the balancing gear parts **320**, the rotational resistance force may be applied to the balancing gear parts **320** through the frictional force generated between the first and second pressing plates **421** and **422** and the balancing gear parts **320**.

[0091] The pressing members **430** are provided between the support plate **410** and the pressing plates **420** and press the pressing plates **420** toward the balancing gear parts **320**. The pressing members **430** according to the embodiment of the present disclosure may be provided in the form of a disc spring, a coil spring, or the like elastically deformable in the longitudinal direction thereof. The pressing members **430** are disposed between the upper surface of the support plate **410** and the lower surface of the second pressing plate **422**. Two opposite ends of each of the pressing members **430** are fixed by adjoining the upper surface of the support plate **410** and the second pressing plate **422**. The pressing member **430** compressed into a predetermined length may be installed between the support plate **410** and the second pressing plate **422**. Therefore, an elastic

restoring force of the pressing members **430** may press the second pressing plate **422** toward the balancing gear parts **320**.

[0092] The stoppers **440** are fixed to the rotation shaft portions **321** and support the support plate **410** on the rotation shaft portions **321**. The stopper **440** according to the embodiment of the present disclosure may be fixed to the rotation shaft portion **321** by means of various coupling structures using press-fitting, screw coupling, or the like. The stoppers **440** fix an installation position of the support plate **410** by supporting the lower surface of the support plate **410** in the state in which the stoppers **440** are fixed to the rotation shaft portions **321**. Therefore, the stoppers **440** may prevent the support plate **410** from being arbitrarily moved in the longitudinal direction of the rotation shaft portions **321** by the reaction force applied from the pressing members **430**.

[0093] The stoppers **440** are installed to be movable in the longitudinal direction of the rotation shaft portions **321**, such that coupling positions of the stoppers **440** with respect to the rotation shaft portions **321** may be adjusted. The stoppers **440** move in the longitudinal direction of the rotation shaft portions **321**, thereby adjusting an interval between the support plate **410** and the pressing plates **420**, more particularly, an interval between the support plate **410** and the second pressing plate **422**. Therefore, the stoppers **440** may change a degree to which the pressing member **430** is compressed, thereby adjusting the magnitude of the pressing force applied to the balancing gear parts **320** from the pressing members **430**.

[0094] Hereinafter, an operational principle of the brake apparatus **1** for a vehicle configured as described above will be described.

[0095] FIGS. **7** to **9** are operational views schematically illustrating a process in which the brake apparatus for a vehicle according to the embodiment of the present disclosure applies the braking force.

[0096] Referring to FIGS. **7** to **9**, to apply the braking force, the electric motor generates power by receiving electric power from the outside, and the connection gear parts **120** and **220** connected to the output shaft of the electric motor through the driving gear rotate.

[0097] The sun gear parts **110** and **210** also rotate in conjunction with the rotations of the connection gear parts **120** and **220**. The planet gears **131** and **231** engaging with the sun gears **111** and **211** revolve about the sun gears **111** and **211** while rotating about their rotation centers.

[0098] As the planet gears **131** and **231** revolve, the carrier parts **150** and **250** coupled to the planet gears **131** and **231** rotate clockwise or counterclockwise. As the carrier parts **150** and **250** rotate, the piston parts **170** and **270** coupled to the carrier parts **150** and **250** move toward the brake pad **20** and come into contact with and press the brake pad **20**.

[0099] Because of several reasons, a larger amount of power provided from the drive part **30** may be transmitted to any one of the pair of pressing parts **100** and **200**.

[0100] For example, when the pressing part **100** disposed at one side (the left side based on FIG. **7**) receives a larger amount of power than the pressing part **200** disposed at the other side (the right side based on FIG. **7**), the piston part **170** disposed at one side may come into contact with the brake pad **20** before the piston part **270** disposed at the other side comes into contact with the brake pad **20**.

[0101] The planet gear part **130** of the pressing part **100** disposed at one side only rotates about its rotation axis in a state in which the piston part **170** disposed at one side is already in contact with the brake pad **20** and the piston part **270** disposed at the other side is not yet in contact with the brake pad **20**. That is, the planet gear part **130** does not revolve.

[0102] More specifically, since the power generated by the operation of the drive part **30** is continuously transmitted to the sun gear **111**, the sun gear **111** continuously rotates. In this case, because the piston part **170** is already in contact with the brake pad **20**, the plurality of planet gears **131** engaging with the sun gear **111** only rotates about their rotation center without revolving.

[0103] The ring gear part **310** disposed at one side rotates clock or counterclockwise in conjunction with the rotations of the plurality of planet gears **131**.

[0104] As the ring gear part **310** disposed at one side rotates, the ring gear outer portion **315** integrally coupled to the ring gear inner portion **311** transmits the rotational force to the balancing gear parts **320**.

[0105] Because of the reaction force between the brake pad **20** and the piston part **170**, a load, which is higher than a load applied when the braking force is released, is transmitted to the ring gear part **310** when the braking force is applied.

[0106] Therefore, the rotational force transmitted to the balancing gear parts **320** from the ring gear part **310** disposed at one side becomes higher than the rotational resistance force applied to the balancing gear parts **320** by the load transmission restricting part **400**, such that the balancing gear parts **320** rotate.

[0107] As the balancing gear parts **320** rotate, the power provided to the pressing part **100** disposed at one side is transmitted to the piston part **270** at the other side sequentially through the balancing gear parts **320**, the outer gear portion **316** disposed at the other side, the inner gear portion **312** of the ring gear inner portion **311**, the planet gear part **230**, and the carrier part **250** coupled to the planet gear part **230**.

[0108] Therefore, the power provided by the drive part **30** is transmitted to the piston part **270** disposed at the other side, which is not yet in contact with the brake pad **20**, and the rectilinear movement of the piston part **170** which is disposed at one side and already in contact with the brake pad **20**, is stopped until the piston part **270** disposed at the other side comes into contact with the brake pad **20**.

[0109] Thereafter, when both the piston parts **170** and **270** disposed at one side and the other side are in contact with the brake pad **20**, the power of the drive part **30** is uniformly transmitted to the piston parts **170** and **270** disposed at one side and the other side. Therefore, the brake pad **20** may be simultaneously pressed by the uniform load.

[0110] FIGS. **10** and **11** are operational views schematically illustrating a process in which the brake apparatus for a vehicle according to the embodiment of the present disclosure releases the braking force.

[0111] To release the braking force, the electric motor generates power by receiving electric power from the outside, and the connection gear parts **120** and **220** connected to the output shaft of the electric motor through the driving gear rotate in a direction opposite to the direction in which the connection gear parts **120** and **220** rotate when the braking force is applied.

[0112] The sun gear parts **110** and **210** also rotate in conjunction with the rotations of the connection gear parts **120** and **220**. The planet gears **131** and **231** engaging with the sun gears **111** and **211** revolve about the sun gears **111** and **211** while rotating about their rotation centers.

[0113] As the planet gears **131** and **231** revolve, the carrier parts **150** and **250** coupled to the planet gears **131** and **231** rotate in a direction opposite to the direction in which the carrier parts **150** and **250** rotate when the braking force is applied. As the carrier parts **150** and **250** rotate, the piston parts **170** and **270** coupled to the carrier parts **150** and **250** move away from the brake pad **20**, thereby releasing the brake pad **20**.

[0114] Because of several reasons, a larger amount of power provided from the drive part **30** may be transmitted to any one of the pair of pressing parts **100** and **200**.

[0115] The reaction force is not applied between the brake pad **20** and the piston part **170** even though a load deviation occurs between the pair of pressing parts **100** and **200** when the braking force is released. Therefore, a load, which is lower than a load applied when the braking force is applied, is transmitted to the ring gear part **310**.

[0116] Therefore, the rotational force transmitted to the balancing gear parts **320** from the ring gear part **310** disposed at one side becomes lower than the rotational resistance force applied to the balancing gear parts **320** by the load transmission restricting part **400**, such that the balancing gear parts **320** do not rotate.

[0117] Because the balancing gear parts **320** do not rotate, the pair of ring gear parts **310** do not

rotate, such that the load is not transmitted through the load transmission part **300**.

[0118] Thereafter, the loads transmitted from the drive part **30** to the pair of pressing parts **100** and **200** are transmitted directly to the pair of piston parts **170** and **270** as they are.

[0119] Therefore, the pair of piston parts **170** and **270** may quickly move away from the brake pad **20**.

[0120] While the present disclosure has been described with reference to the exemplary embodiment depicted in the drawings, the exemplary embodiment is described just for illustration, and those skilled in the art will understand that various modifications of the exemplary embodiment and any other exemplary embodiment equivalent thereto are available. Accordingly, the technical protection scope of the present disclosure should be determined by the appended claims.

[0121] Although exemplary embodiments of the disclosure have been disclosed for illustrative purposes, those skilled in the art will appreciate that various modifications, additions and substitutions are possible, without departing from the scope and spirit of the disclosure as defined in the accompanying claims. Thus, the true technical scope of the disclosure should be defined by the following claims.

Claims

1. A brake apparatus for a vehicle, comprising: a drive part configured to generate driving power; a pair of pressing parts configured to receive the driving power from the drive part and to apply or release a braking force; a load transmission part installed between the pair of pressing parts and configured to transmit a pressing load of any one of the pair of pressing parts to the other pressing part of the pair of pressing parts; and a load transmission restricting part configured to selectively restrict an operation of the load transmission part by pressing the load transmission part, wherein the load transmission part comprises: a pair of ring gear parts connected to each of the pressing parts respectively; and one or more balancing gear parts disposed between the pair of ring gear parts and configured to rotate while engaging with the ring gear parts, and wherein the load transmission restricting part comprises: a support plate fixed to a rotation shaft portion that supports the one or more balancing gear parts so that the one or more balancing gear parts is rotatable; pressing plates disposed to face the support plate and configured to come into contact with the one or more balancing gear parts; and a pressing member disposed between the support plate and the pressing plates and configured to press the pressing plates toward the one or more balancing gear parts.
2. The brake apparatus of claim 1, wherein the load transmission restricting part permits rotation of the one or more balancing gear parts when the pair of pressing parts apply the braking force, and the load transmission restricting part restricts the rotation of the one or more balancing gear parts when the pair of pressing parts release the braking force.
3. The brake apparatus of claim 1, wherein the load transmission restricting part applies a rotational resistance force to the one or more balancing gear parts by pressing the one or more balancing gear parts.
4. The brake apparatus of claim 1, wherein the load transmission restricting part presses the one or more balancing gear parts in an axial direction.
5. The brake apparatus of claim 1, wherein the pressing member is elastically deformable in a longitudinal direction thereof.
6. The brake apparatus of claim 1, wherein the pressing plates comprise: a first pressing plate disposed to face one side of the one or more balancing gear parts; and a second pressing plate disposed to face another side of the one or more balancing gear parts and configured to receive a pressing force from the pressing member.
7. The brake apparatus of claim 6, wherein the second pressing plate is installed to be movable in a

longitudinal direction of the rotation shaft portion.

8. The brake apparatus of claim 1, further comprising: a stopper coupled to the rotation shaft portion and configured to support the support plate.

9. The brake apparatus of claim 8, wherein the stopper is installed to be movable in a longitudinal direction of the rotation shaft portion and is configured to adjust an interval between the support plate and the pressing plates.
